Chemistry Part 7

Historical Development of the Periodic Table

Early 19th Century
Chemists arranged the elements according to their atomic mass led to the development of the first periodic table.

1864
John Newlands (England) observed that by arranging the elements according to their atomic masses, every 8th element have the same properties. This is called the Law of Octaves.

1869
Dmitri Mendeleev (Russia) and Lothar Meyer (Germany) produced a tabulation that resembles the modern day periodic table. Mendeleev's table included 66 elements only.

1913
Henry Moseley (England) proposed that atomic numbers are directly related to the atomic mass

Atomic Number vs. Mass Number

Atomic Number

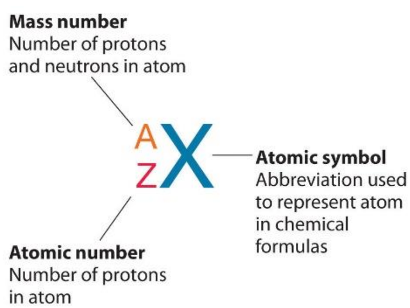
The atomic number is just the number of protons inside the nucleus of an element. This number serves as the identity of an element since no two elements have the same atomic number. Moreover, neutral elements will have the same number of protons and electrons. For these elements, the atomic number can also be used to indicate the number of electrons.

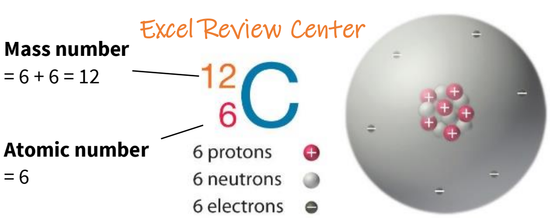
Mass Number

The mass number is the sum of the number of protons and the number of neutrons found inside the nucleus of an atom. To get the number of neutrons, just get the difference between the mass number and the atomic number.

Mass Number = No. of Protons + No. of Neutrons

Mass Number = Atomic No. + No. of Neutrons





Isotope vs Atomic Mass

Atoms of one element can have the same atomic number but different values of atomic weight. These are referred to as **isotopes**. Most elements have two or more isotopes (e.g. Hydrogen has 3 isotopes). Isotopes of one element have the same number of electrons and protons, they only differ in the number of neutrons. Hence, the chemical properties of isotopes on one element must be identical.

Knowing the different isotopes and the mass numbers of the different isotopes of an element, one can now get the atomic mass of the element. The **atomic mass** is the weighted average of the mass numbers of the different isotopes of an element.

Isotopes can either be stable or radioactive. A radioactive isotope will naturally undergo a transformation to an isotope of a different element. For hydrogen, tritium is its radioactive isotope.

Molecules vs. Compounds

Combining atoms of the same element or atoms from different elements in fixed ratios using chemical bonds creates **molecules**. Molecules do not necessarily make compounds, however all compounds are made up of molecules.

Compounds are substances that are made from two or more elements. Examples of molecules made up of atoms from one element only are diatomic molecules or diatomic gases. Examples of these gases are hydrogen (H_2), helium (He_2), oxygen (O_2), nitrogen (N_2), fluorine (F_2), bromine (Br_2), chlorine (Cl_2) and iodine (I_2). Examples of molecules that are made of atoms from more than one type of elements are compounds like water (H_2O) and salt (NaCl).

Types of Compounds

1. **Ionic Compounds** – these are compounds made up of cations and anions (e.g. NaCl is made from Na+ cation and Cl- anion). Most ionic compounds are binary compounds, which means that they are just composed of two types of atoms.

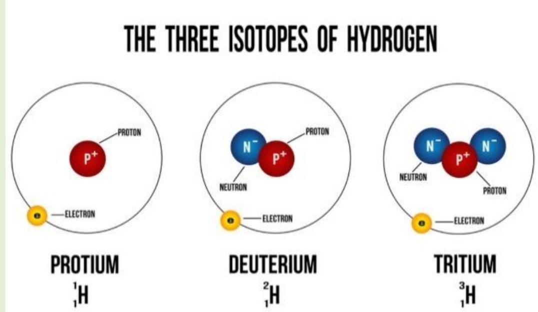
2. **Molecular Compounds** – these are compounds made up of molecules formed from different elements (e.g. H_2O, CO_2). Unlike ionic compounds, these are no longer made up of cations and anions. However, just like ionic compounds, majority of molecular compounds are also binary compounds.

What is an Ion?

When electrons are lost and gained by atoms that are electrically neutral, ions are created. Losing electrons will cause the neutral atom to be positively charged simply because there are more protons than electrons. These positively charged atoms are now called as cations. Gaining electrons, on the other hand, will cause the neutral atom to be negatively charged since there are now more electrons than protons. These negatively charged particles are now called anions. Remember that the number of protons of every atom will never change; only electrons are allowed to move into or away from the atom. Aside from cations and anions, ions can also be classified according to the number of atoms that it consists:

1. Monoatomic ions – these are ions that contain only one type of atom (e.g. Na+, Cl-)

2. Polyatomic ions – these are ions that contain more than one type of atom (e.g OH-, CN+)



Allotropes

If there are two or more forms of one element, then these are called as **allotropes**. For example, O_2 (oxygen gas) is an allotrope of oxygen. O_3 (ozone) is also an allotrope of oxygen.